AI debt and the Treasury term premium: flow or stock?

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The question, in one paragraph. Bond supply can move prices in two ways. A flow effect is the price impact of new supply as it arrives. A stock effect is the lasting impact of how much is outstanding (D'Amico and King, 2013). I measured both for the dollar bonds of the five big AI spenders. There is a flow effect: on the day a deal prices, the 10-year term premium rises about 0.08bp per $1bn of 10-year-equivalent duration, the same sign shows up in another term premium model and in observed yields, and it is gone within days. It is marginal (timing-matched placebo p about 0.08) and does not survive a correction across 216 specifications. The stock effect I cannot measure: 33 months of data cannot tell a stock effect from two series that both trended up. Everything here is exploratory. The one confirmatory test is frozen in code and needs about three more years of deals.

On September 15 the 10-year Treasury yield closed at 5.00% for the first time since July 2007. A week later it hit 5.10%. Around the same time I read a sell-side estimate that roughly 30 basis points of the 2026 selloff came from corporate and mortgage bond supply.

That is a specific, falsifiable claim, and as far as I could tell nobody had published the work behind it.

Meanwhile the five big AI spenders (Amazon, Alphabet, Meta, Microsoft, Oracle) had issued $179.5 billion of dollar bonds in 2026, up from $92.8 billion in 2025 and close to nothing before that. Every one of those bonds is duration that somebody has to hold. If corporate supply moves Treasury yields, this is where you would look.

So I spent the next week or so trying to measure it, using only free data, and then a second pass testing everything the first pass claimed.

The headline result is small and mostly negative. The interesting part is everything that broke on the way there, and specifically the afternoon I spent proving that my own main estimator was incapable of answering the question I had pointed it at.

> *Basis point (bp).* One hundredth of a percentage point. A yield going from 4.94% to 5.00% moved 6bp. Bond people speak entirely in these.
>
> *Term premium.* A 10-year yield is roughly two things added together: what people expect short rates to average over the next ten years, and the extra compensation they demand for locking money up for a decade instead of rolling overnight. The second part is the term premium. It is the part that supply and demand should move, which is why the whole project lives there.

Getting the deals out of EDGAR

There is no free database of corporate bond deals with announcement timestamps. There is EDGAR, which has the filings, and you have to build the database yourself.

A bond offering shows up as three documents, and you need all three because each one is missing something the others have:

- The preliminary prospectus (form 424B) is filed the morning of the deal. It lists the tranches but every dollar amount is blank. Useless for size, perfect for timing.
- The FWP pricing term sheet is filed the same afternoon. It has a literal `Trade Date:` field and the priced sizes.
- The final prospectus lands a day or two later with the authoritative table on the cover: `$5,000,000,000 4.250% NOTES DUE 2031`, one line per tranche.

So: sizes from the final, dates from the FWP. I matched the two by comparing the full set of coupon rates in each, which is a nice check because a bad match shows up as a partial score rather than silently passing. All 16 deals matched at 1.00.

If you skip this and use the final prospectus filing date as your event date, you have introduced a two-day look-ahead into an event study. That is the kind of bug that quietly destroys a result rather than announcing itself.

Here it isn't hypothetical. For every one of the 16 deals, the final prospectus landed after pricing: one business day later for eight of them and two days later for the other eight. Dated by the day each deal priced, the day-0 estimate is +0.081 (Newey-West t 3.47). Dated by the final prospectus filing day instead, it is −0.022 (t −0.69), nothing at all. The effect sits on the day the deal actually priced, and an event study keyed to filing dates would have reported no effect.

Things I only found by actually reading the documents:

- Oracle's February 2026 bond deal came with a mandatory convertible preferred filed alongside it. That part is equity, not a bond, so I kept the bonds and dropped it. Alphabet filed one in June 2026. Neither preferred supplies duration.
- Microsoft's only hit in three years is a debt exchange for Activision paper. Microsoft has issued no new public dollar bonds in the entire window, which surprised me.
- Alphabet's May deal is a yen samurai. Amazon's March deal has a euro twin filed one day after the dollar one. $34.8 billion equivalent of the total is non-dollar, and that supplies duration to bunds and gilts, not Treasuries.
- Amazon's $37B March deal has 11 tranches, two of them floating rate. A floater resets its coupon every quarter, so its interest-rate duration is about three months, not eight years.

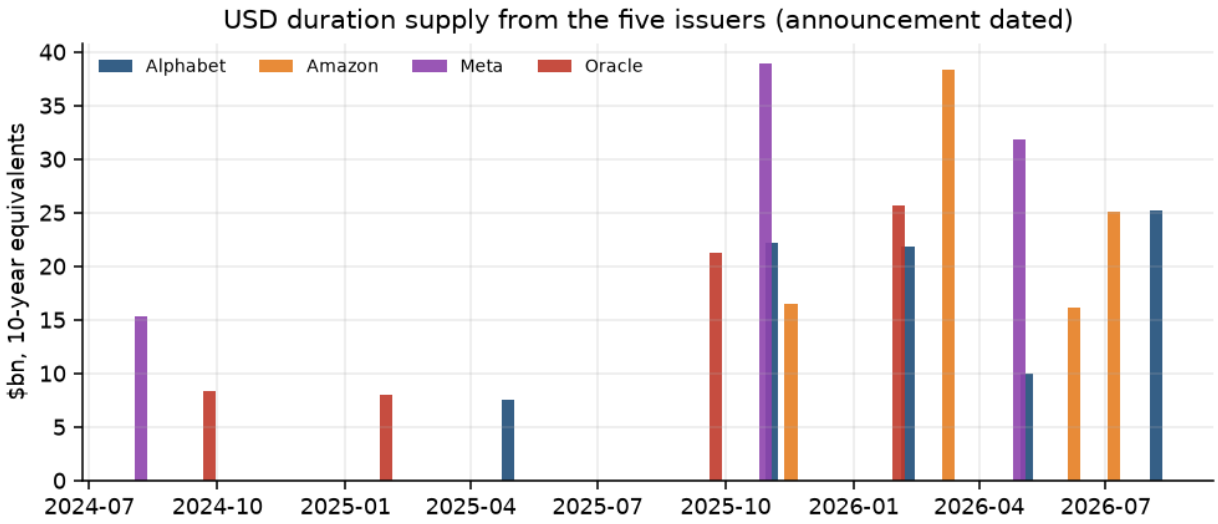
The general point matters more than that example. The whole project is denominated in duration, not dollars.

```
D_t  =  sum over tranches j of  ( notional_j  x  ModDur_j )  /  ModDur_10Y
```

This converts every deal into "how many billions of 10-year notes would carry the same interest rate risk." A $5B 30-year tranche is worth roughly twice a $5B 10-year. The floaters turned out not to matter much here: only 7 of 103 tranches are floating, and treating them as anything from zero to two years of duration moves the main result in the fourth decimal place.

> *Duration.* How much a bond's price moves when yields move. A 10-year note has duration around 8, meaning a 1% rise in yields costs you about 8% of the price. It is the natural unit for "how much interest rate risk is this."

Figure 1. USD duration supply from the five issuers, dated to announcement rather than filing. Nothing before mid-2024, then a step up through 2025, then 2026.



One number gave me confidence the parser was right: my 2025 total came out at $92.75 billion against a widely quoted figure of about $93 billion. My 2026 number, $179.5 billion, is well above the ~$132 billion I have seen quoted, and I think the quoted figure is just older. My own count through early June is $129.5 billion, which matches it. The gap is the two $25 billion deals in July and August.

The first thing that broke

The research design said: drop any deal that prices within three business days of an FOMC meeting, a CPI print, or a payrolls release. You cannot separate a supply shock from a macro shock, so throw those out. Sensible. I wrote it down before touching any yield data.

The event study only uses the jumbo deals, $10 billion or more, which is 12 of the 16. Applied literally, the rule left one of those twelve. The impulse responses later on use all 16.

My first assumption was that the macro calendar is just crowded. It is not crowded enough to explain this. Under a three-day rule, 37% of business days in my sample are clean, so random timing should have kept four or five deals out of twelve. One survived.

Part of the reason is timing. Across all 16 deals, 9 priced within three trading days after a macro release, against 33% of trading days in general (p 0.04). That was one of three windows I tested, though (one, two and three days), and after correcting for that it isn't significant (Holm p 0.13).

Which means the exclusion rule and the thing I was trying to measure are not independent. A symmetric three-day filter around macro events is very close to a filter on corporate issuance itself.

I switched to a narrower rule: drop a deal only if a release lands inside the three-day return window I am actually measuring. That is the version of the concern with teeth, and a market-wide macro shock gets removed by the statistical controls anyway.

Exclusion rule	Events surviving
same day only	11
inside the return window (used)	7
±2 business days	6
±3 business days (as designed)	1

This is a specification change I made after seeing what the original did, and I report it that way everywhere. It is the first of several.

Wait, can this method even work?

The core measurement was supposed to be a "concession": when a company sells a lot of 10-year paper, does the Treasury curve cheapen *specifically* around the 10-year point, beyond whatever the whole curve did that day?

The way you test that is to take the day's change in the yield curve, fit it using only the maturities the deal did not touch, then look at what is left over at the maturities it did touch.

```
dy(tau)  =  a  +  b*f1(tau)  +  c*f2(tau)  +  d*f3(tau)  +  e(tau)
```

where f1, f2 and f3 are the three standard shapes a yield curve moves in (level, slope, curvature, which together explain 99.2% of daily variation). Fit that on the far-away maturities, predict at the deal's maturities, and the residual is your concession.

> *Yield curve factors.* If you look at how the whole curve moves day to day, almost all of it is three patterns: the whole thing shifts up or down (level), the short end and long end move apart (slope), and the middle moves against the ends (curvature). Everything else is noise.

I ran it. The answer was +0.075bp, with a placebo p-value of 0.69. Nothing.

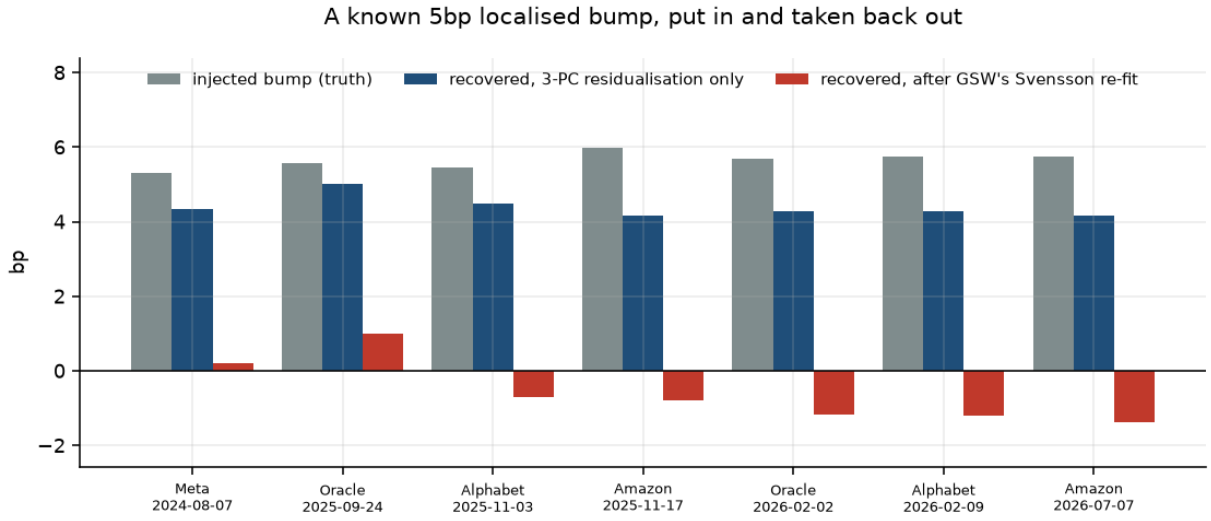
Before writing that up as "no localized effect exists," I looked at the diagnostics. The fit on the control maturities had an R-squared of 1.000. Not 0.999.

That is not a good sign. That is the model explaining the control points perfectly, which means there is no room left for a residual anywhere.

The curve I was using is the Fed's published zero-coupon curve, which is itself a smooth six-parameter function fitted to bond prices. Three factors span almost all of a six-parameter family. So the residual I was hunting for lives below a basis point *by construction*.

Rather than argue about this, I tested it. I took real daily curve moves, injected a known 5bp bump at each deal's maturities, and measured how much of it came back out the other end.

Figure 2. Inject a known 5bp bump, see what the estimator returns. Grey is the truth, blue is after the statistical step, red is after the Fed's own smoothing is applied first.



Stage	How much of the bump survives
The three-factor step alone	77.7%
After the Fed's curve-fitting is applied first	−10.7%

The statistical method is fine. It recovers three quarters of a real bump. The problem is that the published curve is smoothed *before I ever see it*, and that smoothing does not just shrink a local bump. It redistributes it and flips the sign of what is left at the target maturities.

This made a prediction I could check: if the estimate is noise, its sign should wander when I change the control definition. It does. The measured concession swings from −0.167bp to +0.123bp as I move the exclusion band by a couple of years, and both ends are "significant."

So the honest conclusion is not "there is no localized concession." It is "this data cannot detect one, and here is the experiment proving it." Measuring it properly needs transaction-level bond data, which is paid.

That afternoon was the most useful one of the project. Measure what your instrument can see before you trust a null from it.

So the rule from here on: fitted curves, including the Nelson-Siegel fits in the repo, are for descriptive figures only, and no estimate of the supply effect runs on one. The ACM term premium is the one place I lean on a model built from the Fed's fitted curve, which is why the main result is also run on observed yields. I broke this rule once, later, and it cost me.

The flow effect

None of the underlying idea is new. Greenwood and Vayanos (2014) showed Treasury supply moves term premia, and Vayanos and Vila's preferred-habitat model (2021) is the theory for why supply at a maturity should move yields at that maturity. D'Amico and King (2013) split supply effects into flow and stock, which is the frame here. The closest work I found looks at price pressure on the corporate bonds themselves, or at issuers' own rate-lock hedging. I didn't find anyone testing whether this kind of corporate issuance moves Treasury term premia, so that part is new as far as I can tell.

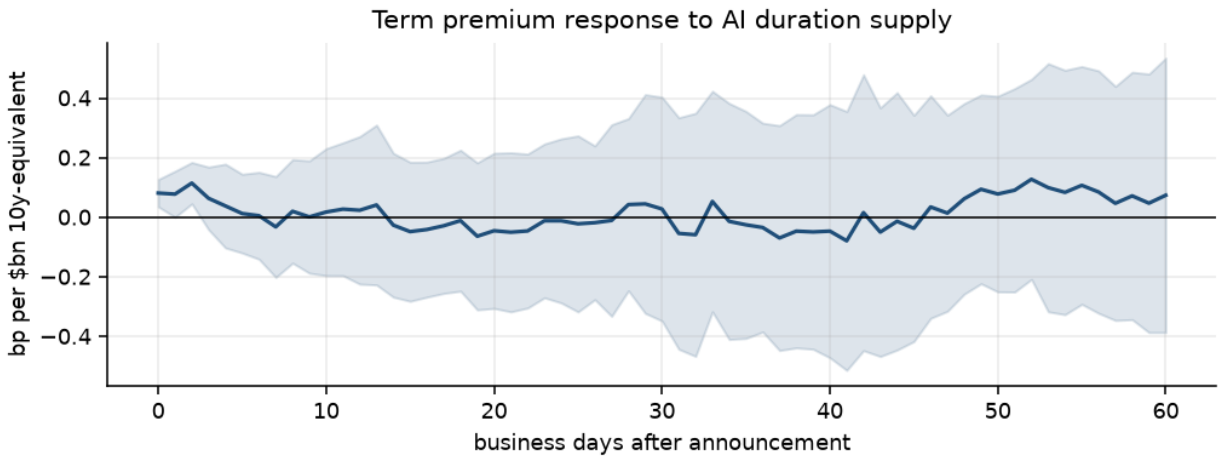
Smooth curves cannot tell you whether a bump is local in maturity. They can tell you whether the long end moved more than the front end and inflation expectations imply, because that compares the same smooth number across event days and normal days, and the smoothing cancels out.

That works. And the more interesting version is to ask how long it lasts, using local projections: a separate regression for each horizon h, from zero to sixty days after the announcement.

```
TP(t+h) − TP(t−1)  =  a_h  +  b_h * shock_t  +  controls  +  error
```

`b_h` is the effect h days later, per billion dollars of 10-year-equivalent duration announced. The controls are the same-day change in the 10-year breakeven, the oil return, the change in the VIX, and the Treasury coupon supply auctioned that day.

Figure 3. The term premium response to announced AI duration supply, with 95% Newey-West bands. Marginal on day 0 under the timing-matched placebo, significant through day 2 under Newey-West, near zero by day 5.



Days after announcement	Effect (bp per $bn)	Timing-matched placebo p	Newey-West t
0	+0.081	0.083	3.47
1	+0.078	0.217	1.97
2	+0.115	0.127	3.24
3	+0.064	0.445	1.18
5	+0.013	0.910	0.19
20	−0.045	not run	−0.34

Visible on impact and gone fast. Under the timing-matched placebo, the inference I trust (it's explained below), it is marginal on day 0 only: p 0.08, then 0.22 on day 1. Under Newey-West, for comparison, it stays significant through day 2. The point estimate is near zero by day 5. Because the per-horizon p-values jump around, I also registered one pooled test before running it: the average response over days 0 to 2 is +0.091, with a matched-placebo p of 0.08, the same as day 0 alone. Scaled by 2026 issuance, $194 billion in 10-year equivalents (the $179.5 billion face value weighted by duration), that is about 16bp of announcement-day effect across the whole year, and a point estimate near zero for what persists.

That doesn't settle the 30bp claim, though. The day-5 estimate implies about 2bp of 2026's supply effect persists, but the placebo band at day 5 is wide: the 95% upper bound is about 41bp (28bp under Newey-West). So this design can't rule out a persistent 30bp effect. Persistence is a question about the stock of debt outstanding, not about announcement days, and it gets its own section below.

Then I tried to break it

I ran an adversarial pass on my own results, then a second, harsher one a few days later, then a third with every new test written down before it ran. Most of what follows came out of those.

The t-statistic was softer than it looked. Newey-West gave me t = 3.47. But plain OLS gives t = 1.84 (p = 0.066), and a randomization test (shuffle the event dates two thousand times, keep the deal sizes, see how often you get a coefficient this big) gives p = 0.078.

The tell was something I should have caught immediately: the Newey-West standard error was smaller than the OLS one, 0.0234 against 0.0441. Newey-West is a correction for correlated errors. When your correction makes your error bars tighter, something is off. Here it isn't the correlation: the errors that matter have almost none. The regressor is zero on all but 16 days, so every robust standard error estimates the variance from those 16 residuals, and they are smaller than average. Equalizing their spread closes about half of the gap between Newey-West and OLS; the other half is unexplained. Whether deal days are really quieter isn't distinguishable from chance either (p 0.13). None of this changes a conclusion, because the primary inference here is the randomization test below, which doesn't estimate a variance at all. Newey-West numbers in this paper are for comparison only.

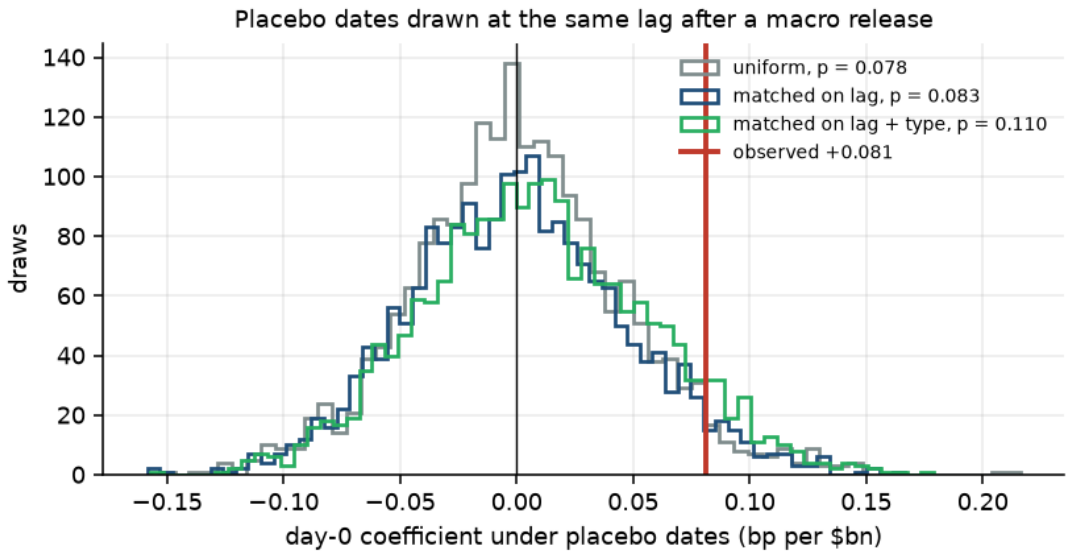
> *Newey-West*. A standard fix that widens your error bars when your data points are correlated over time. Everyone in finance uses it reflexively. It assumes your sample is big and spread out, which mine is not.

The regressor has 679 daily observations, but it is zero on all but 16 of them. The effective sample size is about 9. The small-sample versions of Newey-West that Lazarus, Lewis, Stock and Watson (2018) recommend don't help here, because there is almost no autocorrelation for them to fix: they give t 3.94 and 4.06 (appendix).

The placebo was ignoring the calendar. Shuffling event dates uniformly assumes the real dates are random. They aren't quite. Deals lean toward the days just after macro releases (the 9 of 16 above, marginal as shown). And in this sample the term premium does tend to drift up in those days, about +0.7bp on the release day itself. So maybe my "effect" was just deals inheriting post-release drift.

The fix is to draw every fake date at the same distance after a macro release as the real deal it replaces.

Figure 4. The day-0 coefficient under three kinds of placebo dates, against the real one in red.



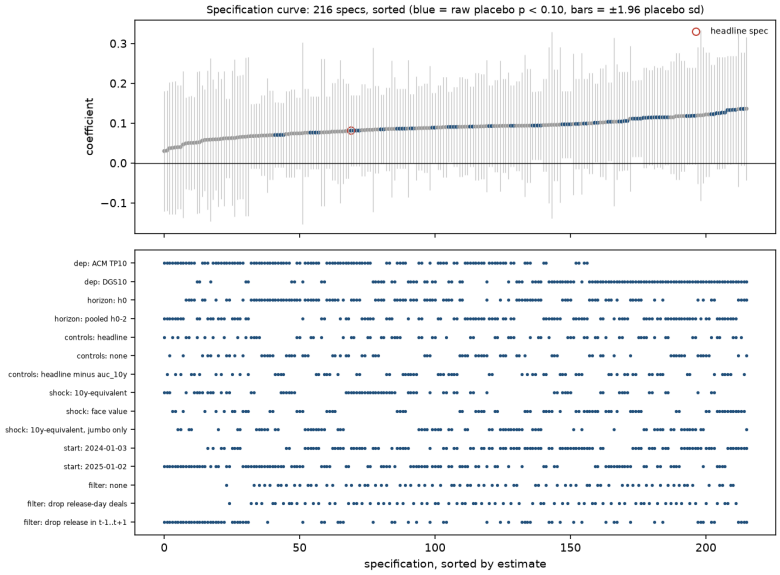
Placebo	Mean	p
uniform	+0.001	0.078
same lag after a release	+0.000	0.083
same lag and same release type	+0.008	0.110

The matched placebo stays centred on zero, so the drift doesn't produce the announcement-day effect. The p-value gets a little worse, which is the honest direction for it to move. I also registered a stricter version that matches each fake date on the volatility of the 20 days before it too. It gives p 0.05 on a sample 20 days shorter. I keep the timing-only placebo as the headline because it was

registered first. It did explain part of something else though: about half of the day-2 "peak" in the impulse response reappears under matched placebo dates, so some of that was drift after FOMC meetings.

One number can be found in 216 ways. The specification has a lot of free choices: which dependent variable, which horizon, which controls, how to size the shock, when the sample starts, and how to handle macro releases. So I registered a family of 216 combinations before running any of them, and ran every one against the same placebo draws.

Figure 5. Every specification, sorted by its estimate. Bars are ±1.96 placebo standard deviations; blue points have a raw placebo p below 0.10. The panel shows which choices produced each estimate. The headline spec is circled.



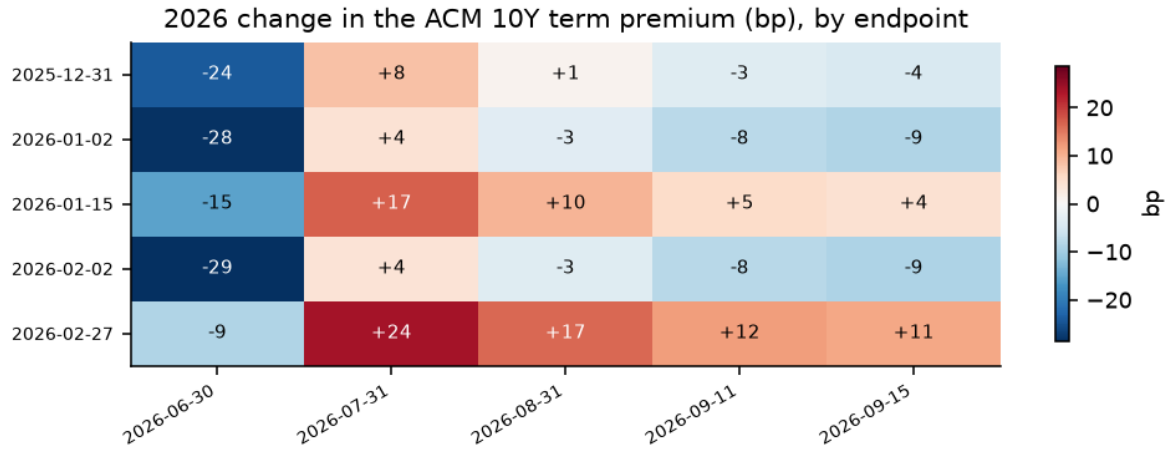
All 216 estimates are positive. 38% have a raw p below 0.10 and 9% below 0.05. After correcting across the family, with Romano-Wolf stepdown p-values and Benjamini-Hochberg, none survives at 0.10. The headline spec's Romano-Wolf p is 0.69. So the sign is very stable and the significance is not: this is a consistent small positive number, not a discovery.

It survives the other usual suspects. Weighting by volatility, scaling by it, and robust regressions that discount outliers all give +0.070 to +0.081. None of the ten most influential days in the sample is a deal day; the top of that list is the August 2024 yen carry unwind and the April 2025 tariff selloff. The controls are not collinear. The full set of checks is in the appendix.

It is also as much as this sample can see. With 16 deals, the smallest day-0 effect this design detects with 80% power is 0.116, above the estimate itself. Power against the estimate is about 54%. At 32 deals, the held-out test's gate, it would be 79%.

The endpoint result was wrong. I had written that the 2026 term premium fell 8.6bp while expected short rates rose 79bp, so the selloff could not be a supply story. Then I varied the start and end dates.

Figure 6. The 2026 change in the term premium by start and end date, in bp.



It ranges from −29bp to +24bp and is positive in 12 of the 25 windows. Moving the end date forward one week, past the Fed's 16 September hike, takes it from −8.6bp to −21.9bp. And over the full 2024 to 2026 sample the 10-year rose 101bp of which +90bp is term premium. The exact opposite of the calendar-2026 cut I had picked.

So a claim I had already written into a document was an artifact of the window I chose. I kept the original text and added a correction rather than quietly editing it.

What survived. The core coefficient held up. Dropping any one of the 16 deals leaves it between +0.068 and +0.094, and dropping all three pairs of deals that overlap gives +0.091. The p-value I trust is the timing-matched placebo's, about 0.08, because it asks the right question: are these dates special compared with dates the same distance after a macro release? A wild cluster bootstrap gives 0.015, but it only asks whether the coefficient is noise within these 16 windows, which is a narrower question. So this is the one result I still stand behind, at p around 0.08 rather than the 0.0005 I originally thought, and with the specification curve's warning attached.

Is it just the ACM model?

The ACM term premium is a model estimate, and 96% of its daily moves are explained by five shapes of the same yield curve. So on its own it can't tell a premium move from any long-end move. I ran the identical day-0 regression on the other public term premium model, Kim and Wright's, and on observed Treasury yields that no model touches.

Dependent variable	Day-0 effect (bp per $bn)	Timing-matched placebo p
ACM term premium	+0.081	0.070
Kim-Wright term premium	+0.038	0.048
2-year yield	+0.058	0.254
5-year yield	+0.076	0.112
7-year yield	+0.093	0.051
10-year yield	+0.097	0.025
20-year yield	+0.106	0.014
30-year yield	+0.115	0.007
10s minus 2s	+0.040	0.279

The ACM p here (0.070) differs from 0.083 only because these runs share one set of placebo draws across all the dependent variables.

Kim-Wright agrees in sign at about half the size. Kim-Wright is also an affine model, so agreement is support, not proof. The observed yields are the model-free check, and they are where the effect is clearest: it rises with maturity, from not significant at 2 years to p 0.007 at 30. That is ten dependent variables, so the smallest p-values here should be read with that count in mind.

Is it where the deals were?

Preferred-habitat theory says supply at a maturity should move yields at that maturity. Fitted curves can't see that, as the injection experiment showed, so this uses observed yields only. I split each tranche's duration between the two nearest Treasury maturities, so every deal has an exposure profile across the 2, 5, 7, 10, 20 and 30-year points. 54% of all the deals' duration sits at the 30-year point and 18% at the 10-year.

Then the test: on each day, do the maturities that got more of the deal move more than the other maturities that same day? Comparing within the day removes any move of the whole curve. The answer is +0.103 per $bn of exposure, with a placebo p of 0.17. Positive, and not distinguishable from a curve-wide move. So I can't say the effect is localized at the issued maturities. A curve-wide move fits a hedging or risk-sentiment story as well as a preferred-habitat one.

The stock question

The flow effect is what happens on the day. The stock effect is whether the total amount outstanding raises the term premium for as long as it is outstanding. The 30bp claim is really a stock claim.

I built the cumulative outstanding AI duration, with no decay, and compared it with the term premium month by month: 33 months. Both series need care, because a level regression of one trending series on another finds a relationship whether or not one exists.

The two standard tests for whether the levels move together disagree: Engle-Granger says yes (p 0.043), Johansen says no (5.10 against a 15.49 cutoff). By the rule I registered, either test was enough, so the levels regression ran: +0.21bp per $bn outstanding, with a Newey-West t of 3.87 and R^2 0.41. I don't read that as a stock effect. The stock is a near-straight ramp, the term premium also rose over the sample, and a ramp regressed on a rising series is exactly the kind of number this project keeps learning to distrust. It goes on the list of things that looked too good.

The version that doesn't depend on the levels is monthly changes on monthly new issuance: −0.067 per $bn, permutation p 0.71, with a minimum detectable effect of 0.50. That is no evidence either way. The stock question is open, and 33 months can't close it.

The satisfying part

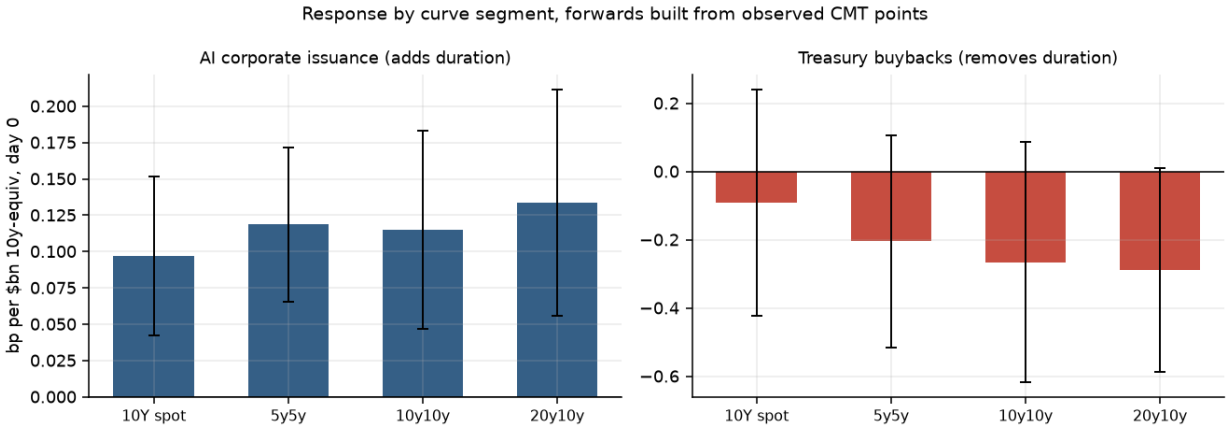
If this is really about duration absorption, then somebody *removing* long duration should move the same numbers the other way. Treasury does exactly that: it runs buyback operations in the same market, at daily frequency, and in September it doubled the size of its 10-to-30-year buybacks.

So I ran the identical regression on Treasury's long-end buyback operations, with the prediction written down first: the coefficient should be negative. It is, −0.20 per $bn.

There's a catch I missed the first time. Buyback dates and maximum sizes are announced in advance, and three quarters of long-end operations fill exactly to that maximum, so only about 12% of what I was counting was actually news on the day. Using just the surprise part, how much more or less got filled than expected, the sign holds: −1.45 per $bn two days later (t −2.46), on 18 operations. That's about 18 times the issuance coefficient, in the same units, but the surprises are tiny, around $1 billion on average against about $21 billion for a typical AI deal. So I trust the sign more than the size. The earlier version isn't useless either. Lou, Yan and Zhang (2013) showed Treasury prices sag before auctions and recover after even though everyone knows the auction is coming, because dealers can only hold so much. Anticipated flows still move prices.

Weak Treasury auctions push the same way. When bid-to-cover comes in low, the term premium rises (t 1.81 across all auctions). That one is consistent with the Lou, Yan and Zhang result, though at t 1.81 I wouldn't call it a replication.

Figure 7. Day-0 response by curve segment, forwards built from observed Treasury yields. Issuance adds duration and pushes yields up; buybacks remove it and push them down.



Three flows, three signs that line up. That sounds better than it is. If there were no effect at all, each sign would be a coin flip, and three matching coin flips happens one time in eight. It's consistent. It isn't proof.

I thought I'd ruled out hedging. I hadn't.

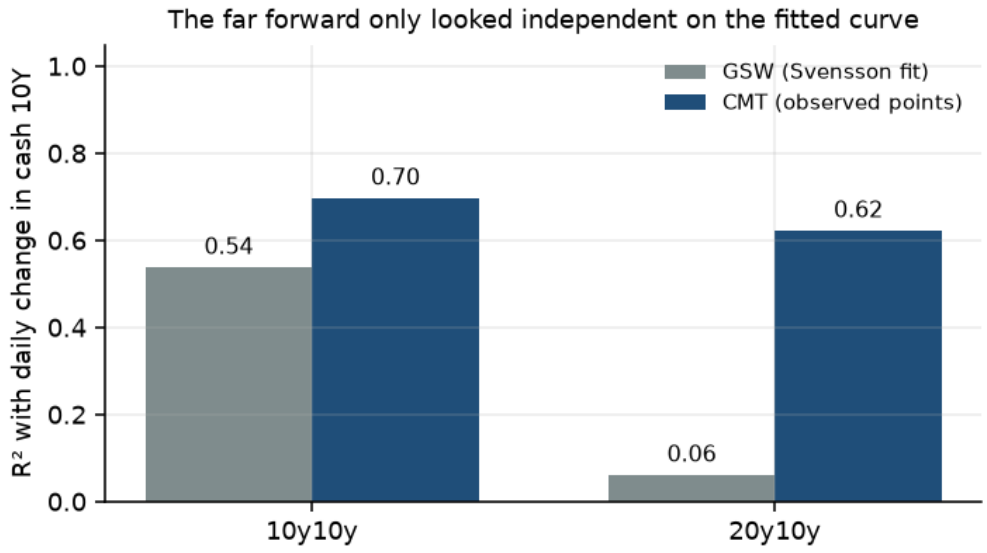
The obvious objection to all of this: when a company sells $25 billion of bonds, the banks and investors hedge by selling Treasuries, mostly around the 10-year point. That flow pushes 10-year yields up for a day or two and then unwinds. No change in what investors actually demand for holding duration. Just plumbing.

My first answer was the 20y10y forward, the 10-year rate implied twenty years out. On the Fed's fitted curve it barely moved with the cash 10-year at all (R^2 0.06), and the effect still showed up there. So, I argued, it can't be a hedge sitting on the 10-year.

> *Forward rate.* The 20y10y is the market's implied 10-year yield starting twenty years from now. You back it out of today's curve.

That argument was wrong, and for exactly the reason from the injection experiment. The fitted curve has very few bonds past twenty years pinning it down, so its far forward is mostly the fit's own noise. Build the same forward from actual observed Treasury yields and it moves with the 10-year a lot.

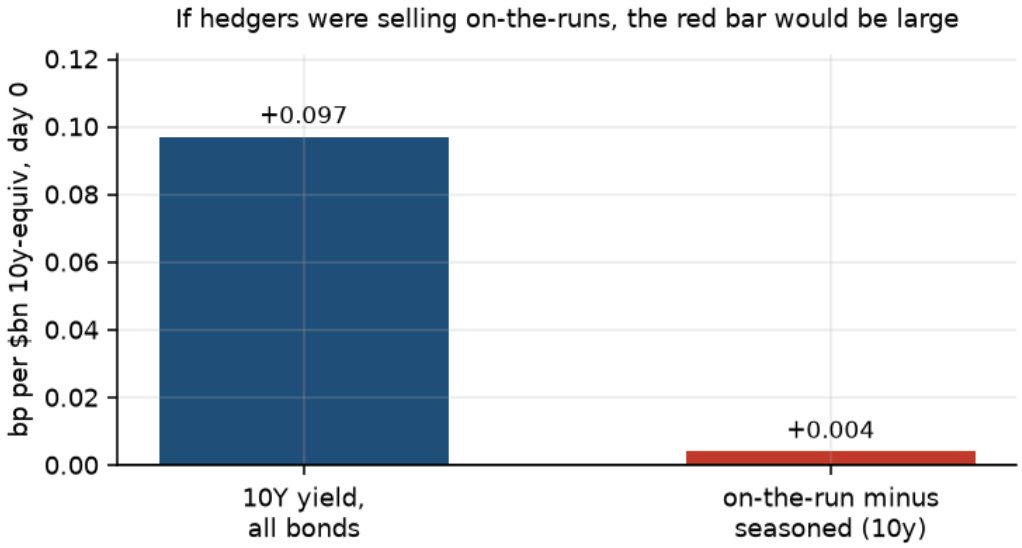
Figure 8. How tied each forward is to the cash 10-year, on the fitted curve versus observed yields.



R² of 0.62, not 0.06. The two versions of "the 20y10y forward" only correlate with each other at 0.38. I had learned that exact lesson with the injection experiment and then walked straight back into it.

So I needed a test that uses observed prices. There's a nice one hiding in plain sight. The Treasury's constant-maturity yields are built from on-the-run bonds, the newest and most liquid issues, which are what hedgers actually sell. The Fed's fitted curve deliberately excludes on-the-runs and uses only older, seasoned bonds. So the gap between the two at 10 years measures how cheap on-the-runs are relative to everything else. If hedgers were dumping on-the-runs, that gap should widen on deal days.

Figure 9. The day-0 move in the 10-year yield, next to the move in the on-the-run minus seasoned spread.



It barely moves: 4% of the yield move. Whatever happened on deal days, it happened to on-the-run and seasoned bonds together. Real yields tell a similar story. A hedge in nominal Treasuries should push nominal yields more than inflation-protected ones, but real yields move 0.87 times as much as nominals, and breakevens, the gap between them, barely move (+0.015bp per $bn, t 0.8).

That rules out the simplest version of the hedging story. It doesn't rule out all of it. Hedgers can use futures instead of cash bonds, and the bond a 10-year future delivers is usually a seasoned one that the fitted curve also sees. Closing that gap needs swap rates or intraday prices, and I don't have either for free.

Was the supply expected?

If markets see a deal coming, the part they expected should already be in prices, and only the surprise should move on the day. So I tried to forecast each issuer's next 90 days of issuance from its own 10-K and 10-Q passages on liquidity and capital spending, and split every deal into the forecast part and the surprise.

The text didn't help. After removing near-duplicate passages, 1,831 became 729. Scored out of sample on 75 filings, the plain baseline, each issuer's issuance over the past year, had the lowest error (166.6). Adding a count of the filing's passages (173.9), the capital spending figures it states (174.5), or a ridge regression on sentence embeddings of the text (212.3) all did worse. So the forecast I used is the baseline, and the text stays out of the result.

With that forecast, most of every deal counts as surprise. The surprise part comes out at +0.110 (placebo p 0.097) and the expected part at −0.012 (p 0.93). That is the predicted pattern, but the difference between them isn't significant (p 0.49), so this can't tell anticipated from surprise supply.

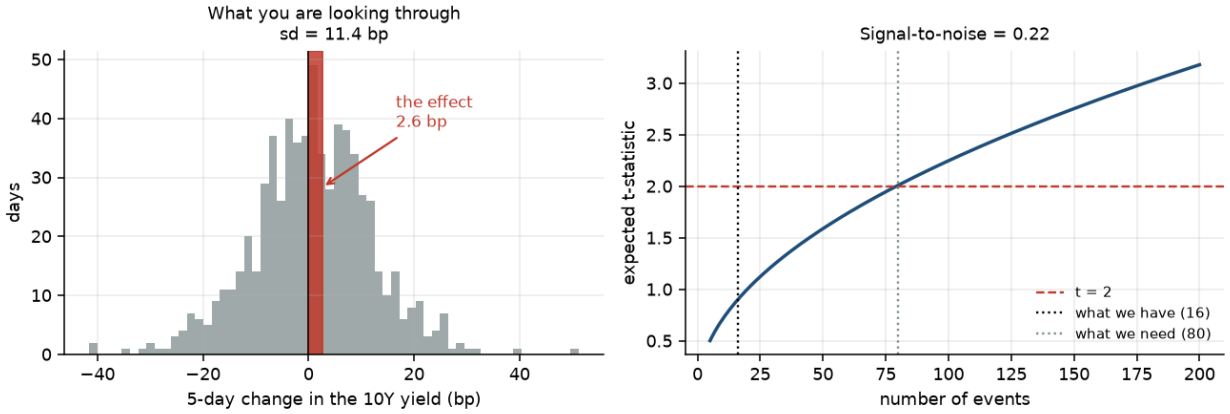
Can you trade it?

This is where the obvious question comes in: if the effect is real, why do the backtest numbers jump around so much?

The first version of the trade was: buy a 10-year note at the close of announcement day, sell five days later, capture the reversion. In-sample Sharpe 0.87. Then everything fell apart under inspection.

The position is barely about the deal. A five-day 10-year position has a yield standard deviation of 11.4bp, which is 91bp of price. My realized P&L standard deviation was 62bp, lower mostly by luck: the 10-year happened to move less than usual across these 16 windows (7.8bp sd). For five days you own whatever CPI, the Fed and oil do. The deal is incidental.

Figure 10. Left: the distribution of 5-day moves in the 10Y, with the expected effect marked in red. Right: how many events you need before that effect becomes visible.



The signal is a quarter the size of the noise. A mean-size deal moves the 10-year about 2.56bp. Five-day noise is 11.4bp. Signal-to-noise is 0.22. To see that at t = 2 you need about 80 events. I have 16.

I also found a real bug. Deal size explained 1.2% of the outright P&L. Hedge out the 2-year and it explains 21.1%, seventeen times more. The issuance shock is a *steepening* shock, not a level shift. I had tested the hypothesis with a control for the 2-year and then traded it without one, which is just sloppy.

The result depends on how you build the trade. Same 16 events, same idea, three ways of expressing it:

Version	Sharpe
outright 10Y, 5 days	+0.87
size-scaled 2s10s steepener	−1.10
size-demeaned spread	+0.27

A two-point Sharpe swing from construction choices alone. Holding period does the same thing: −0.88 at two days, +0.87 at five, −0.69 at ten, with no pattern.

By the end I had run 30 distinct configurations against the same 16 events. At a signal-to-noise of 0.22, thirty configurations will likely produce at least one Sharpe above 1 by luck. Whichever one looks best is the one that got luckiest, and no amount of principled-sounding story afterwards changes that.

Forecasting says the same thing. A cleaner version of the question: does knowing about supply improve a forecast of the term premium one or three days ahead, compared with a forecast that only uses the usual market variables? I refit every day on past data only and scored each forecast on days it hadn't seen. Adding supply made the forecast slightly worse (out-of-sample R^2 −0.061 against −0.051

at one day), and the test for whether it adds anything gives p 0.86. Even the market-variables forecast loses to just predicting the historical average. Nothing here beats the simplest models, so I stopped there instead of reaching for anything fancier.

The volatility is not a bug to engineer away. It is the honest width of the uncertainty around a 2.6bp effect measured sixteen times.

What would settle it

- **Wait for the held-out test.** Every p-value above comes from searching over specifications on the same data, and no after-the-fact correction survives that. So I froze one test in code, hashed it, and it only runs on deals announced after 24 September. It needs 32 of them for decent power, which is about three years at this year's pace. That's fewer than the 80 in Figure 10 because it's a different test: a one-day move with controls, which is much less noisy than a five-day trade.
- **Drop the AI framing.** Run every jumbo investment-grade deal, not five names. That is hundreds of events instead of sixteen, and it is the only change that actually fixes the power problem. The AI angle is what makes the question interesting and also what makes it unanswerable.
- **Get intraday.** The effect is a same-day phenomenon measured on daily closes. Most of what there is to see is happening inside the announcement day.
- **Transaction-level bond data.** The one thing that would let me answer the original maturity-localization question at all.

What this can't answer

- **Power.** At 16 deals the design detects day-0 effects of 0.116 or more with 80% power, and the estimate is 0.081. A true effect of this size would be missed about half the time.
- **Exploratory status.** Several hundred test statistics were computed on this sample, and every new one since the first draft is listed in the repo's test registry. Only the frozen held-out test is confirmatory.
- **Model dependence.** The headline uses one term premium model. The other model agrees in sign at half the size, and observed yields agree, but the premium itself is never observed.
- **Mortgage supply.** The sell-side claim was about corporate and mortgage supply together, and there is no free daily mortgage issuance series. As a rough bound, I added the weekly change in the Fed's mortgage holdings, which fell $518 billion over the sample, to a weekly version of the regression. The AI coefficient moved by 4.9% and the mortgage term wasn't significant (p 0.24). That is a bound, not a control.
- **Timing.** Greenwood, Hanson and Stein (2010) show companies time long-dated issuance around Treasury supply, which would make the shock depend on the outcome. Two checks so far. Deals priced after calmer 20-day stretches of term premium volatility than comparable dates (p 0.015), though not after lower VIX. And a logit of deal days on conditions known the day before (an auction that day, the past week's 10-year move, the curve slope, the Baa credit spread) is marginal

jointly (permutation p 0.07), driven by the slope: deals are more likely when the curve is steeper (Wald p 0.04). Both issuance and the slope rose over the sample, and this test doesn't separate that from day-to-day timing. Its power against an odds ratio of 2 per standard deviation is 39%. Neither check can rule timing out at 16 deals.

Why this was fun

The transferable lesson is not about bonds.

Seven separate times in this project I built something that produced a confident, significant-looking number, and every time the number came from the measurement rather than the market:

- The concession estimator ran on the Fed's fitted curve, which is already smooth, so it fit with an R^2 of 1.000.
- The ACM term premium is 96% explained by the shape of the curve, so on its own it can't tell a premium move from any long-end move. That's why the main result is also checked on observed yields.
- A Japan hedged-yield test had the 10-year Treasury yield inside the regressor, which is how it got a t-statistic of 8.2 with the wrong sign.
- An auction tail was measured against the previous close, so any day yields rose looked like a bad auction.
- The far forward on the fitted curve is mostly the fit's own noise past twenty years, which made it look independent of the 10-year.
- A TIPS regression controlled for breakevens, which are just nominal minus real, so real and nominal matched to four decimal places.
- The stock regression put a ramp on a rising series and got t 3.87.

Each time the tell was a number that looked too good.

The far forward stung, because I'd already written the lesson down.

The useful habit I came away with is to spend an afternoon testing whether your measurement can detect a thing you put there on purpose, before you spend a week interpreting what it says about the real world. The injection experiment cost almost nothing and it was the difference between publishing "there is no effect" and publishing "this method cannot see one."

The second habit is to let the randomization test win. Analytical standard errors are assumptions wearing a lab coat. Shuffling your event dates a few thousand times is not.

The third is to write the prediction down before the test runs. I got several wrong, and they are in the repo next to the ones I got right.

Code, data and the full audit are at github.com/aaravraina3/ai-duration-supply. Everything is from free sources: FRED, the Fed Board, the New York Fed, SEC EDGAR, BLS and Treasury Fiscal Data. The `research/` folder has the hypothesis log with statuses, the preregistrations with every prediction, and the test registry.

Appendix: diagnostics

Day-0 local projection, all from `output/tables/diagnostics_*.csv`.

Standard errors (t on the shock):

Method	t
OLS	1.84
Newey-West, 0 lags (heteroskedasticity only)	3.25
Newey-West, 5 lags (used)	3.47
Newey-West, 20 lags	3.77
Newey-West, automatic bandwidth (11 lags)	3.56
Equal-weighted cosine, t critical values	3.94
Bartlett with fixed-b critical values	4.06
HC3	3.03

Corrected estimates, against placebo dates matched on timing and pre-deal volatility:

Estimator	Effect	Placebo p
OLS	+0.081	0.053
Term premium change divided by trailing volatility	+0.025 (in sd units)	0.059
Weighted by inverse volatility	+0.070	0.100
Huber robust regression	+0.080	0.039
Median regression	+0.077	0.053

Other checks. Residuals are serially correlated (Ljung-Box p 0.0003 at one lag), heteroskedastic (White p below 0.0001), and fat-tailed (kurtosis 6.3). Variance inflation factors run from 1.01 to 1.27. The term premium level and the decayed supply stock are nonstationary, so every specification that put a level on the right-hand side of a daily change is retired. A break test at the start of 2025 rejects (p 0.0002), but only 2 deals come before the break.